UNIVERSITY OF THE PEOPLE

CS 2204-01 Communications and Networking -

AY2026-T4

Assignment Activity Unit 2

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Date: April 18, 2026

Data Communication and Network Infrastructure Design

Introduction

The rapid growth of industrial and commercial networking demands thoughtful selection of transmission media, encoding schemes, and multiplexing technologies. In a scenario where a startup company must simultaneously support a manufacturing plant burdened with heavy electromagnetic interference (EMI) and an office complex requiring robust high-speed data services, these choices become critically important. This paper addresses three core questions: which line coding scheme best mitigates EMI in the factory environment, which transmission media suit different segments of the office network, and which multiplexing technique most

effectively serves the company's diverse customer base. Each answer is grounded in a comparative analysis of available technologies and supported by established networking literature.

Question 1: Line Coding Schemes and EMI Mitigation

Overview of Line Coding

Line coding is the process of converting digital data (binary 1s and 0s) into digital signals suitable for physical transmission. The choice of encoding scheme has a direct impact on a system's ability to maintain clock synchronization, resist noise, and recover data accurately all of which are critical in electrically noisy environments such as a manufacturing plant (Dordal, 2023). The three schemes under comparison are Non-Return to Zero (NRZ), Manchester Encoding, and Differential Manchester Encoding.

Non-Return to Zero (NRZ)

NRZ is the simplest encoding method. In NRZ-L (Level), a binary 1 is represented by a high voltage and a binary 0 by a low voltage. In NRZ-I (Inverted), the signal transitions on a 1 and remains steady on a 0. While NRZ offers simplicity and bandwidth efficiency requiring only one signal change per bit, it has a critical weakness: long sequences of identical bits produce no transitions. This means the receiver can lose synchronization, a problem known as clock drift (Dordal, 2023). In an EMI-heavy manufacturing environment, where electrical noise from heavy machinery can corrupt signals and cause long flat lines in the waveform, NRZ is the least reliable choice. Any prolonged period of identical bits can cause the receiver to miscount bits entirely.

Manchester Encoding

Manchester Encoding resolves the synchronization problem inherent in NRZ by guaranteeing a transition in the middle of every bit period. A binary 1 is represented by a low-to-high transition, and a binary 0 by a high-to-low transition. This mid-bit transition allows the clock signal to be embedded within the data stream itself, eliminating the need for a separate synchronization channel (Dordal, 2023). As a result, Manchester Encoding is self-synchronizing: even under noisy conditions, the receiver can continuously realign its clock with each incoming bit. This is a significant advantage in factory environments.

However, Manchester Encoding doubles the signal bandwidth requirement compared to NRZ, since it requires two signal changes per bit. This means that at a given baud rate, the effective

bit rate is halved. Despite this trade-off, the gain in reliability under interference is well worth the cost in industrial deployments (Forouzan, 2022).

Differential Manchester Encoding

Differential Manchester Encoding builds standard Manchester Encoding by changing the rule for interpreting bits. While the mid-bit transition is always present (ensuring synchronization), the presence or absence of a transition at the start of each bit interval determines the bit value: a transition at the start indicates a 0, and no transition indicates a 1. This scheme is inherently more robust because it does not rely on the absolute voltage level of the signal only on whether a transition occurred. This makes Differential Manchester especially resistant to polarity reversal and signal inversion, which can occur due to physical changes in cabling or EMI events (Dordal, 2023).

In practice, Differential Manchester is used in token ring networks and is specified in IEEE standards for reliable LAN communication. Its noise resistance and self-synchronizing nature make it the superior choice for high-interference environments.

Comparison Table

Feature	NRZ	Manchester	Differential Manchester
Synchronization	Poor (no guaranteed transitions)	Good (mid-bit transition)	Excellent (always mid-bit transition)
EMI Resistance	Low	Moderate	High
Bandwidth Efficiency	High	Low (2x bandwidth)	Low (2x bandwidth)
Complexity	Simple	Moderate	Higher
Polarity Sensitivity	Yes	Yes	No

Recommended Scheme for the Manufacturing Plant

For the manufacturing plant, Differential Manchester Encoding is the most appropriate choice. The heavy machinery in the plant creates persistent electromagnetic interference that can corrupt signal voltage levels and even reverse signal polarity in poorly shielded cable runs. Differential Manchester's reliance on transitions rather than absolute voltage levels makes it robust to these disturbances. Its guaranteed mid-bit transition ensures clock synchronization is maintained

even when the signal is degraded by noise. While the doubled bandwidth usage is a trade-off, in a factory LAN operating at standard Fast Ethernet speeds (100 Mbps), this is entirely manageable with modern hardware. Reliability and data integrity in a safety-critical manufacturing environment outweigh bandwidth efficiency concerns.

Question 2: Transmission Media for the Office Network

Overview

The selection of transmission media profoundly affects the performance, scalability, and cost of a network. For the office complex supporting video conferencing, cloud applications, and high-speed internet, the three primary wired options are twisted pair cable, coaxial cable, and fiber-optic cable. Each serves different roles within a layered network architecture (Forouzan, 2022).

Twisted Pair Cable

Twisted pair cable consists of pairs of copper wires twisted together to reduce electromagnetic crosstalk and interference. Unshielded Twisted Pair (UTP) is the most widely deployed LAN medium and is used in Ethernet standards including Cat5e (up to 1 Gbps over 100 meters) and Cat6/Cat6a (up to 10 Gbps over shorter distances). It is inexpensive, easy to install, and widely compatible with standard networking equipment.

However, twisted pairs are susceptible to EMI, have limited bandwidth compared to fiber, and support relatively short maximum distances before signal degradation. In a standard office setting with distances of under 100 meters per segment, these limitations are not prohibitive. Twisted pair is the clear choice for workstation connections it is cost-effective at scale, and its distance limitations pose no issue within a single floor or building (Dordal, 2023).

Coaxial Cable

Coaxial cable uses a central copper conductor surrounded by a dielectric insulator, a braided shield, and an outer jacket. This construction gives it better noise resistance than UTP and supports longer distances and higher bandwidth. Historically used in Ethernet (10BASE2, 10BASE5) and still common in cable TV and broadband internet distribution, coaxial cable occupies a middle ground between twisted pair and fiber.

In modern enterprise networks, coaxial is rarely used for LAN infrastructure due to its bulk, higher cost per meter, and the fact that fiber has largely replaced it for backbone roles. However, in scenarios where the office network must connect to external service providers via cable internet (DOCSIS), coaxial cable remains relevant for the last-mile segment.

Fiber-Optic Cable

Fiber-optic cable transmits data as pulses of light through a glass or plastic core. It supports extraordinarily high bandwidths (multi-gigabit to terabit speeds), extremely long transmission distances (several kilometers for single-mode fiber), and is completely immune to electromagnetic interference. These properties make fiber the backbone medium of the modern internet and all high-capacity enterprise networks (Dordal, 2023).

The trade-off costs: fiber cable, transceivers, and installation labor are significantly more expensive than copper alternatives. However, backbone connections where the cable carries aggregated traffic from many workstations, the cost per bit is fully justified. Fiber is also the only realistic medium for connecting the office to external ISPs over distances exceeding a few hundred meters.

Comparison Table

Criteria	Twisted Pair	Coaxial Cable	Fiber Optic
Bandwidth	Up to 10 Gbps (Cat6a)	Up to ~1 Gbps (DOCSIS)	Terabits per second
Max Distance	~100 m (UTP)	Several kilometers	Tens of kilometers
EMI Susceptibility	Moderate	Low	None (immune)
Cost	Low	Moderate	High
Best Use	Workstation access layer	ISP last mile / CATV	Backbone & ISP uplinks

Recommended Media Allocation for the Office Network

For connecting individual workstations to access-layer switches, Cat6 UTP twisted pair is the optimal choice. It provides sufficient bandwidth for video conferencing and cloud applications at low cost and is easy to install in typical office cabling runs under 100 meters. For the backbone connections between floors and between buildings within the office complex, where aggregated traffic demand is highest, single-mode or multimode fiber-optic cable is recommended. Fiber ensures that the backbone is never a bottleneck as the organization grows. For linking the office to

external ISPs, fiber-optic uplinks (such as those provided over SONET/OTN or Ethernet over fiber standards) are the standard in commercial settings, offering the gigabit-speed connectivity required for cloud services and supporting future bandwidth growth (Dordal, 2023). Coaxial cable may be appropriate only if a cable-based ISP service (DOCSIS) is the only available broadband option.

Question 3: Multiplexing Techniques for the Customer Base

Overview of Multiplexing

Multiplexing is the technique by which multiple communication signals are combined and transmitted simultaneously over a single shared channel, with the signals separated at the receiving end. For a company offering internet services to residential and business customers over a shared infrastructure, selecting the right multiplexing approach determines how efficiently bandwidth is allocated, how well the system scales, and how resistant it is to interference (Forouzan, 2022).

Time Division Multiplexing (TDM)

In TDM, the available transmission time is divided into fixed slots, and each subscriber is assigned one or more slots in a repeating cycle. Every user transmits in their designated time slot, regardless of whether they have data to send. TDM is inherently synchronous: all participating devices must be coordinated to the same precise clock a requirement that is fully met in fiber-based systems like T-carrier lines and SONET (Dordal, 2023).

TDM is highly effective for applications with constant, predictable traffic such as voice telephony, where each DS0 channel requires exactly 64 kbps at all times. The SONET hierarchy (STS-1 through STS-768) is built on TDM principles and supports line rates from 51.84 Mbps up to nearly 40 Gbps over optical fiber. However, TDM is inefficient when traffic is bursty: if a user has no data in their slot, that bandwidth is wasted and cannot be reallocated to other users. This makes TDM less suitable for modern internet traffic, which is inherently variable in nature.

Frequency Division Multiplexing (FDM)

In FDM, the available frequency spectrum of the channel is divided into non-overlapping sub-bands (sub-channels), each assigned to a different user or signal. All users transmit simultaneously, but on different carrier frequencies. The classic example is radio broadcasting multiple stations coexist on the same physical medium (the atmosphere) because each occupies a

distinct frequency band. In wired systems, cable TV uses FDM to carry hundreds of channels over a single coaxial cable. In fiber-optic networks, the optical equivalent Wavelength Division Multiplexing (WDM), allows many independent data streams to travel on the same fiber, each on a different light wavelength (Dordal, 2023).

FDM is well-suited to continuous, fixed-rate applications and scenarios with diverse service types occupying well-defined frequency bands. However, it requires guard bands between sub-channels to prevent interference, which wastes some spectrum. It is also less flexible than modern packet-based approaches when demand patterns shift dynamically.

Comparison Table

Feature	TDM	FDM
Bandwidth Allocation	Fixed time slots per user	Fixed frequency bands per user
Scalability	Moderate (requires re-slotting)	Moderate (limited by spectrum)
Efficiency for Bursty Traffic	Low (idle slots wasted)	Moderate
Interference Resistance	High (digital, synchronized)	Moderate (guard bands needed)
Typical Use Case	Voice networks, SONET/T-carrier	Radio, cable TV, WDM fiber

Recommended Multiplexing Approach

For the company's network upgrade intended to serve a diverse customer base of residential users and businesses, a hybrid approach is most effective in practice. The backbone infrastructure connecting the company to upstream providers and carrying aggregated traffic should use WDM (Wavelength Division Multiplexing) over fiber-optic cables. WDM supports terabit-scale capacity by multiplexing dozens or hundreds of wavelengths on a single fiber and is the backbone technology of modern internet infrastructure (Dordal, 2023).

For the access network, the portion directly serving customers, TDM remains highly relevant for voice services and dedicated leased lines to business customers who require guaranteed, constant bandwidth. For residential internet and variable-demand business customers, modern broadband systems such as DOCSIS (for cable) or GPON (for fiber-to-the-home) use

dynamic allocation schemes that combine TDM principles with statistical multiplexing, automatically assigning capacity to users based on current demand. This hybrid model delivers the reliability of TDM for guaranteed services and the efficiency of dynamic allocation for bursty internet traffic, optimizing bandwidth utilization across the full customer base.

Conclusion

In designing the data communication and network infrastructure for this startup manufacturing and office complex, each technology choice must be matched to its specific operational context. Differential Manchester Encoding is selected for the manufacturing plant because its transition-based bit representation provides the synchronization stability and noise resistance required in a high-EMI environment. In the office network, twisted pair of copper serves the access layer efficiently and cost-effectively, while fiber-optic cable is essential for backbone and ISP interconnection links where bandwidth and distance demands are greatest. For the customer-facing network infrastructure, WDM over fiber provides the capacity backbone, while TDM and dynamic multiplexing serve voice and broadband subscribers respectively. Together, these choices create a network that is reliable, scalable, and fit for purpose across all operational environments.

References

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